CSE 4304-Data Structures Lab. Winter 2021
Date: August 10, 2022
Target Group: All
Topic: Strings, sorting, binary search

Instructions:

- Task naming format: fullID_T01L02_2A.c/cpp
- If you find any issues in problem description/test cases, comment in the google classroom.
- If you find any test case that is tricky that I didn't include but others might forget to handle, please comment! I'll be happy to add.
- Use appropriate comments in your code. This will help you to easily recall the solution in the future.
- Obtained marks will vary based on the efficiency of the solution.
- Do not use <bits/stdc++.h> library.
- Modified sections will be marked with BLUE color.

Group	**Tasks**
2B	1 2 5 6
2A	2 3 7 8
1A/1B	1 6 7 8

Task-01:

Alexander wants to fight for the Coding Club. In each round, there will be ***S*** soldiers with various powers. There will be ***R*** rounds to fight and in each round, his power will change. With power ***P***, he can kill all the soldiers whose power is less than or equal. The sum of powers of the soldiers will be considered as the total point for that round. After each round, all the soldiers who are dead in the previous round will be reborn. So in each round, there will be ***N*** soldiers to fight.

Your job is to count the number of soldiers that he can kill in each round and provide the total points that Alexander has achieved.

Input will start with ***N*** & ***R*** followed by ***N*** power levels of the soldiers. The next lines will contain the ***R*** power levels of Alexander for the different rounds.

Input	**Output**
7 3 2 3 1 6 5 7 4 3 10 2	3 6 7 28 2 3
10 8 42 12 25 13 5 55 43 41 17 35 2 39 13 22 73 29 0 35	0 0 6 107 3 30 4 47 10 288 5 72 0 0 6 107

Note:

- Try to optimize the time-complexity of your code as much as possible.
- Do not use any built in function.

Task-02:

Suppose, you are playing the 'Bulls and Cows' game with your friend.
You write down a secret number and ask your friend to guess what the number is. When your friend makes a guess, you provide a hint with the following info:

- The number of 'bulls': which are digits in the guess that are in the correct position.
- The number of 'cows': which are digits in the guess that is in your secret number but are located in the wrong position. Specifically, the non-bull digits in the guess that could be rearranged such that they become bulls.

Given the secret number `secret` and your friend's guess, return *the hint for your friend's guess*. The hint should be formatted as `"xByC"`, where x is the number of bulls and `y` is the number of cows. Note that both `secret` and `guess` may contain duplicate digits.

The first line of input will contain the secret and the following line will be the guess. The secret and guess consist of only digits and have equal lengths.

Input	**Output**	**Explanation** (Bulls are connected with a '|' and cows are underlined)
1807 7810	1B3C	1807 | 7810
1123 0111	1B1C	1123 | 011 or 011 (Note that only one of the two unmatched 1s is counted as a cow since the non-bull digits can only be rearranged to allow one 1 to be a bull)
1123 0145	1B0C	1123 | 0145
2441139 1234522	0B4C	2441139 4322 or 4322 or 4322
2441139 4444111	3B1C	2441139 || | 4444111 or 4444111

		(There is only one ‘1’ in the secret, don’t confuse with the two ‘1’s in the guess)

Bulls and Cows wiki: https://en.wikipedia.org/wiki/Bulls_and_Cows
Note: 10 bonus marks if you can bring an O(N) solution.

Task-03:

Timmy starts his new job as a DJ in dubstep nation. In dubstep nations, the DJs are supposed to create a new type of music mix made with the word ‘dub’. The process is like, every original song must start with the word “dub” and after every word of the original lyric the word “dub” should be added. But Timmy does not like this music. So he decided to get the original song lyrics from the mixed dubstep music. Your task is to help Timmy get the original song lyrics from dubstep.

The first line of input contains the total length of the dubstep song. The second line contains the dubstep song lyric.

Input	**Output**
37 dubneverdubgonnadubgivedubyoudubup	never gonna give you up
47 dubhisdubpalmsdubaredubsweatydubkneesdubweakdub	his palms are sweaty knees weak

Task 5:

Suppose, you are a professional robber planning to rob houses along a street. Each house has a certain amount of money stashed, the only constraint stopping you from robbing each of them is that adjacent houses have security systems connected and it will automatically contact the police if two adjacent houses were broken into on the same night.

(Consecutive houses have a unique amount of money)

Given an integer array nums representing the amount of money of each house, return the maximum amount of money you can rob tonight without alerting the police. (Take input until -1 is given)

Input	**Output**	**Clarification** (Picked houses)
1 2 3 1 -1	4	1 3
2 7 9 3 1 -1	12	2 9 1
2 1 1 2 -1	4	2 2
1 2 7 2 1 9 -1	17	1 7 9

Task 06:
Two arrays are given, one containing the positions and another containing the values. Your job is to rearrange the values based on the positions.

The first line contains the position values. Keep taking input in the first line until -1 is given. The following line will contain the values corresponding to those positions.

Input	**Output**
3 1 2 -1 32 54.7 -2	54.7 -2 32
5 1 6 4 3 2 -1 3.75 3.88 3.75 3.92 3.99 3.84	3.88 3.84 3.99 3.92 3.75 3.75

Note: **05 Bonus marks for O(N) solution.**

Task 7:
Given your total obtained marks in the final exam (out of 100), your task is to return the obtained grade.

The first line of input will contain two numbers ‘level’ and ‘trials’. Then each line will contain a number and the associated grade level.
After the levels are inserted, the trails will be inserted. Each trial will contain an integer value.
For each trial, you have to print the corresponding grade.

Input	**Output**
J	F D B+ A+ B
7 9 39 F 40 D 50 C 60 B 70 A- 80 A+ 90 A* 39 35 0 100 89 77 60 61 59	F F F A* A+ A- B B C

Note: Try to find the most efficient solution.

Task 08:

One measure of ‘undersortedness’ in a sequence is the number of pairs or entries that are out of order with respect to each other. For instance, in the letter sequence ‘DAABEC’, the measure is 5, since D is greater than four letters to its right and E is greater than one letter to its right. This measure is called the number of inversions in the sequence. The sequence ‘AACEDGG’ has only one inversion (E and D) and it is nearly sorted. On the other hand, the sequence ‘ZWQM’ has 6 inversions; it is as unsorted as can be, exactly in the reverse order.

You are responsible for cataloging a sequence of DNA strings (sequences containing only the four letters A, C, G, and T). However, you want to catalog them not in alphabetical order, but in the order of ‘sortedness’- from the ‘most sorted’ to ‘least sorted’. All the strings are of the same length.

The first line of input will contain two numbers ‘length’ and ‘n’ representing the length of each sequence and the total number of strings.

Your task is to arrange the string from ‘most sorted’ to ‘least sorted’ order. On the right of each string, show the degree of unsortedness.

Input	**Output**
10 6 AACATGAAGG TTTTGGCCAA TTTGGCCAAA GATCAGATTT CCCGGGGGGA ATCGATGCAT	CCCGGGGGGA 9 AACATGAAGG 10 GATCAGATTT 11 ATCGATGCAT 17 TTTTGGCCAA 36 TTTGGCCAAA 37